

Response to Reviewers

Spatial Modeling of Tissues for Morphogenic Field Analysis

PLOS ONE, PONE-D-26-04349 (Osgood-Zimmerman & Raredon)

Draft of 2026-07-27 | Revision deadline: 2026-08-07

INTERNAL DRAFT, NOT FOR SUBMISSION.

Items marked **[TODO: ...]** are not yet done. Statuses: **[DONE]** the work is finished and verified; **[DRAFTED]** response text is written but the corresponding manuscript edit has not been made; **[TODO]** not started. Set `\draftmodefalse` and confirm no `\todo` remains before submitting.

Dear Dr. D'Acquisto and Reviewers,

Thank you for the careful and constructive reviews. We are grateful for the time invested, and we think the manuscript is substantially better for the revision.

Reviewer 2 asked whether the framework identifies biologically meaningful spatial structure beyond smoothing, increased feature dimensionality, or clustering. We regarded this as the central question of the review, and we have addressed it directly with two new controls (Section R2.8): a coordinate shuffle with full model refitting, and a concordance test against the independent anatomical annotations published with the source dataset. The coordinate-shuffle control is the more informative of the two, and it is unambiguous: refitting the identical pipeline on spatially permuted data collapses the emergent domains entirely. We also report, in the same section, a third control that we ran and chose not to present, together with our reasons.

We have also moderated the generalizability and biological-interpretation language throughout, clarified the statistical model and its implementation, and added a supplementary table mapping every mathematical object in the manuscript to the code that implements it.

Reviewer comments are reproduced verbatim in italics, followed by our response. Changes are marked in the accompanying *Revised Manuscript with Track Changes*.

Sincerely,

Aaron Osgood-Zimmerman and Micha Sam Brickman Raredon

1. Status summary

Internal only, delete this section before submission.

Item	Ask	Status
JR-1	PLOS style and file naming	Done
JR-2	Role of funder statement	Needs author input
JR-3	Remove funding text from Acknowledgments	To do
JR-4	PLOS L ^A T _E X template	Done

Item	Ask	Status
JR-5	Upload Supplemental Materials as SI	To do
JR-6	Evaluate recommended citations	Drafted (see R2.10)
R1.1	Comparative analysis vs existing methods	Drafted (see R2.8, R2.10)
R1.2	Tissue sample and methods detail	To do
R1.3	Figures too small, reorganize	To do
R1.4	Small-scale disease proof of concept	Drafted (respectful pushback)
R1.5	Better graphical abstract	To do
R2.1	Define model parameters and priors at first use	Drafted
R2.2	Notation-to-code supplementary table	Drafted, table not built
R2.3	Clarify that final analyses used observed nUMI	Drafted
R2.4	Define metrics; Table 1 abbreviations and percentages	To do
R2.5	Rationale for Cd44 as representative feature	Needs author input
R2.6	Elongated-pixel distortion in Figs 3 and 4	To do
R2.7	Clearer captions for Figs 5 and 6	To do
R2.8	Controls and baselines	Done, two re-reported, one withdrawn
R2.9	Reproducibility documentation	Partly done
R2.10	Positioning vs related methods	Drafted
R2.11	Moderate generalizability claims	Drafted
R2.12	Cautious biological language	Drafted

2. Journal requirements

JR-1. PLOS ONE style requirements and file naming

[DONE]

■ *Please ensure that your manuscript meets PLOS ONE's style requirements, including those for file naming.*

The manuscript has been reformatted to the PLOS ONE style templates. Section headings, title capitalization, the reference style, and file naming now follow the PLOS ONE guidelines. See also JR-4.

JR-2. Role of the funder

[NEEDS AUTHOR INPUT]

■ *Please state what role the funders took in the study. If the funders had no role, please state: "The funders had no role in study design, data collection and analysis, decision to publish, or preparation of the manuscript."*

We confirm the following statement, which we have included in our cover letter for the journal to enter into the online submission form:

The funders had no role in study design, data collection and analysis, decision to publish, or preparation of the manuscript.

[TODO: Both authors must confirm this is accurate for T32GM086287 (NIGMS) and for the Yale and Bucknell startup funds before it goes in the cover letter.]

JR-3. Funding text in the Acknowledgments

[TODO]

We note that you have provided funding information that is not currently declared in your Funding Statement. However, funding information should not appear in the Acknowledgments section or other areas of your manuscript.

The funding text has been removed from the Acknowledgments section of the manuscript. The Acknowledgments now contain only non-financial acknowledgments. The funding information appears solely in the Funding Statement, and our updated statement is reproduced in the cover letter.

[TODO: Actually remove the funding sentence from the Acknowledgments in paper-plos.tex, around line 376, and check that no funding language survives elsewhere in the text.]

JR-4. PLOS L^AT_EX template

[DONE]

Please update your submission to use the PLOS LaTeX template.

The manuscript has been ported from the Elsevier `elsarticle` class to the PLOS L^AT_EX template, including the PLOS preamble, title and author block, and the `plos2025` bibliography style.

JR-5. Supporting Information upload

[TODO]

Please upload a copy of Supporting Information Figure/Table/etc. “Supplemental Materials” which you refer to in your text on page 19.

The supplemental materials, previously available only through external repository links, are now uploaded as Supporting Information files S1 through S8 and are cited individually at the appropriate points in the text.

[TODO: Package the eight Supplements for Publication directories as PLOS SI files, confirm each S-number in the manuscript matches the uploaded file, and check the SI captions render.]

JR-6. Recommended citations

[DRAFTED]

If the reviewer comments include a recommendation to cite specific previously published works, please review and evaluate these publications to determine whether they are relevant and should be cited.

We evaluated every method named by Reviewer 2 (SpatialDE, BOOST-GP, Spatial Variance Component Analysis, and SpatialDM) and judged all of them relevant. Each is now cited and discussed in a new positioning subsection; see our response to R2.10. We have additionally cited CytoSignal

(Liu et al., *Nature Genetics*, 2026), concurrent work that was not available at the time of our original submission and that analyzes the same source dataset.

3. Academic editor

It seems to me that the Reviewers have asked a number of clarifications and more detailed explanation for the approaches used in the study while finding the overall idea of interest. I would suggest the authors to fully address the comments made by the Reviewers as this might significantly improve the impact of the study and its overall take-home message.

We agree, and we have treated every reviewer comment as actionable. The clarifications requested by Reviewer 2 concerning the statistical model, its implementation, and the evaluation metrics were fair; the manuscript understated how much detail a reader needs to reproduce or critically assess the work. The point-by-point responses below describe each change.

4. Reviewer 1

R1.1. Comparative analysis with existing spatial modeling techniques

[DRAFTED]

Recommend the authors to perform a comparative analysis with other existing techniques for spatial modelling techniques as verifications.

We have added comparison and validation along three axes, described in full in our response to Reviewer 2’s comment R2.8.

First, within our own framework we compare candidate convolution operators quantitatively, and this comparison now includes the product operator used by CytoSignal. The common-minimum operator we adopt yields more spatially coherent domains than the product operator on this dataset.

Second, we compare the emergent domains against an external reference that we did not produce: the anatomical annotations published with the source dataset by Chen et al. (2022). This is a comparison to an independently derived, expression-based segmentation of the same tissue section.

Third, we added a spatial-shuffle null control with full model refitting, which is the standard validation device in this literature.

We want to be straightforward about what we have not done. We have not run a full method-versus-method benchmark against SpatialDE, BOOST-GP, SVCA, SpatialDM, or CytoSignal. Those methods answer different questions from ours: they test individual features or interactions for spatial significance, whereas our contribution is a continuous-field representation with quantified uncertainty and the clustering of the full high-dimensional signaling field. There is no shared target quantity on which a head-to-head accuracy comparison would be meaningful without first defining a benchmark task, which we regard as its own piece of work. We have instead positioned our approach against each of these methods conceptually and explicitly (R2.10), and we now state this limitation in the Discussion rather than leaving it implicit.

R1.2. Detail on the tissue sample and methods

[TODO]

■ *The methods on the tissue/patient sample obtained here is missing and lacks details.*

We have expanded the dataset description in Materials and methods. The revised text specifies the source publication and accession, the developmental stage and section identity, the platform and its capture chemistry, the physical dimensions of the field of view analyzed, the binning applied, the number of spatial locations and features modeled, and the filtering that produced the analyzed feature set.

[TODO: Write this expansion into the Dataset selection and Dataset preprocessing subsections. Concrete facts to state: Chen et al. 2022 Stereo-seq mouse embryo (MOSTA), section E16.5.E1S3; field of view x from 15000 to 25000 and y from 11500 to 18000 in raw chip coordinates; bin size 50; 26000 spatial locations; 918 curated features of which 912 produced converged fits; FANTOM5 ligand-receptor pairings, of which 1477 have both partners among the converged features. Note that the sample is a mouse embryo from a public atlas, not a patient sample, which is what the reviewer may be assuming.]

R1.3. Figure size and organization

[TODO]

■ *The data presented in the figure is on the small side of the spectrum, recommend the authors to reorganise them.*

We have regenerated the figures at larger physical size and higher resolution, and reorganized several panels so that the comparisons a reader is asked to make are adjacent. Figures 3 and 4 have also been corrected for the aspect-ratio distortion noted by Reviewer 2 (R2.6).

[TODO: Regenerate all figures. Coordinate with R2.6 (aspect ratio) and R2.7 (captions) so each figure is touched once.]

R1.4. Small-scale disease proof of concept

[DRAFTED, PUSHBACK]

■ *Since the author claims the spatial modelling provides multidimensional information to guide disease assessment, it might be good for the author to use this paper to perform a small scale POC work.*

We appreciate this suggestion and have thought about it carefully, but we have concluded that adding a disease dataset is not the right change for this manuscript, and we have instead removed the framing that invited it.

The manuscript did not report a disease result; disease was raised only as future motivation. On reflection, that framing promised more than the work delivers, which is also the substance of Reviewer 2's comment R2.11 about generalizability. Rather than adding a second dataset, we have moderated the disease language so the stated scope matches the evidence: this is a methodological proof of concept on healthy developing tissue.

We think this is the more honest resolution. A disease proof of concept done properly requires matched healthy and diseased tissue, attention to the confounding between disease state and tissue composition, and enough replication to distinguish a disease effect from between-sample variation. Done in the time available on a single additional section, it would produce an illustrative figure that we could not defend statistically, which is the opposite of what this manuscript argues for. We

have noted the application to diseased tissue as clearly future work in the Discussion, alongside the lung-tissue direction already mentioned there.

R1.5. Graphical abstract

[TODO]

Minor comment: The graphical abstract submitted did not do the work justice, consider a better images rather than copy-pasting from the data section.

We agree, and we have replaced the graphical abstract with a purpose-designed schematic rather than panels reused from the figures.

[TODO: Design and produce the new graphical abstract. Raredon.]

5. Reviewer 2

We thank Reviewer 2 for an unusually thorough and useful review. The comments identified real gaps, particularly the absence of controls, and the manuscript is materially stronger as a result.

R2.1. Precision of the statistical model specification

[DRAFTED]

First, the statistical model is not specified with sufficient precision. The manuscript introduces a hierarchical statistical framework, but several key parameters are not clearly defined, including θ_i , θ_N , $\Sigma(\theta_i)$, $\Sigma(\theta_N)$, the covariance function, prior structure, and the relationship between the abstract mathematical model and the computational implementation. Since this model is central to the manuscript, all parameters should be introduced clearly at first use.

We accept this criticism. The model section was written at a level of abstraction that made the framework readable but left the actual fitted model underdetermined. We have rewritten it to define every symbol at first use and to state the concrete choices used in the reported analyses.

Specifically, the revised Materials and methods now state that the spatial random effect for feature i is a zero-mean Gaussian process with a Matérn covariance function, represented through the stochastic partial differential equation approach, so that $\theta_i = (\rho_i, \sigma_i)$ collects the range and marginal standard deviation for feature i , and $\Sigma(\theta_i)$ denotes the resulting Matérn covariance matrix evaluated at the mesh nodes. The smoothness parameter is fixed rather than estimated ($\alpha = 2$, giving Matérn smoothness $\nu = 1$ in two dimensions). We place penalized-complexity priors on θ_i : the range prior satisfies $P(\rho_i < 500) = 0.95$ and the marginal standard deviation prior satisfies $P(\sigma_i > 0.1) = 0.05$, with the range in the same physical units as the chip coordinates. A sum-to-zero constraint is applied to the field for identifiability against the intercept. The symbols θ_N and $\Sigma(\theta_N)$ played a role only in the latent total-count model, which the final analyses did not use; see R2.3, where we have clarified their status rather than leaving them ambiguous.

We have also added an explicit paragraph connecting the abstract model to its implementation, and a supplementary table making the mapping object by object (R2.2).

[TODO: Write this into the “A general spatial statistical model for spatialomics” subsection of paper-plos.tex. The prior values above are VERIFIED against the canonical pipeline (supp-4-lik-comparison/02-array-run-model.R line 63, inside the Poisson branch: `matern.pri.feat.count <- c(500, .95, .1, .05)` passed to `inla.spde2.pcmatern`

as `prior.range = c(500, 0.95)`, `prior.sigma = c(0.1, 0.05)`, with `alpha = 2` and `constr = TRUE`). Note the file-level default at line 32 is `c(500, .95, 1, .05)`, with `sigma 1` rather than `0.1`; it is overridden inside the Poisson branch, so quote the line-63 values. Posterior summaries use `n.samp <- 1000` draws.]

R2.2. Notation-to-code mapping

[DRAFTED, TABLE NOT BUILT]

Second, the correspondence between the notation in the manuscript and the public code repository is difficult to follow. Readers should be able to identify where mathematical objects such as $\lambda_i(s)$, $F_i(s)$, $N(s)$, α_i , $\Sigma(\theta_i)$, and θ_i are implemented in the code. I recommend adding a supplementary table mapping each manuscript term to its corresponding code object, script, or model component.

We have added this table as Supporting Information. Each row gives a mathematical symbol, its meaning, the script in which it is constructed, and the name of the corresponding code object, so that a reader can move from any equation in the manuscript to the lines that implement it.

[TODO: Build the table. One row per symbol, minimally: $\lambda_i(s)$, $F_i(s)$, $N(s)$, α_i , θ_i , $\Sigma(\theta_i)$, θ_N , $\Sigma(\theta_N)$, the mesh, the Matérn SPDE object, the likelihood, and the posterior predictive draws. Anchor each row to a script and object name in Supplements for Publication/code/spatial-modeling/. Verify every path and object name against the actual files rather than from memory.]

R2.3. Treatment of total counts $N(s)$

[DRAFTED]

Third, the treatment of total counts $N(s)$ requires clarification. The manuscript presents a latent model for $N(s)$, but later states that modeling $N(s)$ did not substantially alter the results and that observed $n\text{UMI}$ values were used directly. The authors should clearly state whether the final reported analyses used a joint latent model, an observed offset or exposure term, or a simplified model using observed $n\text{UMI}$ values.

The reviewer is right that this was ambiguous, and we have stated it plainly.

All analyses reported in the figures and in Table 1 use the observed number of unique molecular identifiers at each location as a fixed offset, that is, as an exposure term entering the Poisson mean multiplicatively and treated as known rather than modeled. No result in the manuscript comes from the joint latent model for $N(s)$.

We retain the latent formulation in the model development section because it is the general form of the framework and because we tested it, but we now flag its status explicitly at first presentation, rather than only in passing later. We also state what the comparison showed: fitting the joint latent model for $N(s)$ did not materially change the estimated feature fields, which is why the simpler and computationally cheaper offset specification was adopted for the full feature set. The supporting comparison is the analysis in Supplement 3.

[TODO: Add the clarifying sentence at the point where the latent $N(s)$ model is first introduced, not only in the Results. Confirm the Supplement 3 comparison supports the “did not materially change” wording, and quantify it if the numbers are available.]

R2.4. Prediction metrics and model comparison

[TODO]

Fourth, the prediction metrics and model comparison need clearer explanation. The manuscript compares Poisson, ZIP, and ZAP models using MAE, RMSE, Dawid-Sebastiani score, and log score, but the definitions, orientation, and interpretation of these metrics are not sufficiently clear. Table 1 should define all abbreviations and explain the percentages in parentheses. The authors should also clarify which criteria were prioritized and why the evidence supports selecting the Poisson model over ZIP and ZAP.

We have added explicit definitions for each metric, stated the orientation of each, expanded the Table 1 caption to define every abbreviation and the parenthetical percentages, and stated which criteria we prioritized and why.

In summary: mean absolute error (MAE) and root mean squared error (RMSE) are point-forecast accuracy measures; the Dawid-Sebastiani score (DS) and logarithmic score (LS) are proper scoring rules that assess the full predictive distribution rather than a point summary. All four are negatively oriented, so smaller values indicate better performance. We prioritized the proper scoring rules, because the manuscript’s argument concerns the quality of the predictive distribution and its uncertainty rather than point accuracy alone.

[TODO: Still needs the actual justification for Poisson over ZIP and ZAP written against the Table 1 numbers, including the metrics that disagree (MDS favours ZAP at 82% and ZIP at 85%, while MAE, RMSE and MLG favour Poisson), and the metric definitions written as equations in the Methods. Two things about Table 1 are now verified from 03-array-post-process.R and can be stated directly. (i) The parenthetical percentages are the mean over features of the ratio of that model’s score to the Poisson score *for the same feature*, expressed as a percentage, so 100% means indistinguishable from Poisson. They are NOT the ratio of the two averages printed in the table, which is why 0.29 against 0.21 shows as 111% rather than 138%. The current caption calls this an “average percentage difference,” which is a misnomer for a ratio and should be reworded. (ii) The averages exclude features with abnormal values indicating nonconvergence, via the filter $MAE < 100 \ \& \ abs(MDS) < 100$, which the code comments say drops 4 and 3 features respectively. The caption’s “all 918 modeled ligand/receptor features” is wrong on both counts and needs restating.]

R2.5. Rationale for Cd44 as the representative feature

[NEEDS AUTHOR INPUT]

| *For example, the rationale for selecting Cd44 as a representative example should be explained.*

We have added the rationale to the text and to the figure caption.

[TODO: Needs the authors’ actual reason. Candidate justifications, to be confirmed rather than assumed: Cd44 sits at a typical expression level and sparsity for the modeled feature set, so it is representative rather than favourable; it is broadly expressed across several anatomical regions, so the raw-versus-normalized contrast is visible; it is a well characterized surface receptor. State the real reason, and if the feature was chosen for visual clarity, say so and note that the full per-feature results are in the supplement.]

R2.6. Elongated-pixel distortion in Figures 3 and 4

[TODO]

Figures 3 and 4 appear visually distorted due to elongated rectangular pixels, which should be corrected or clearly identified as a visualization artifact.

We have regenerated Figures 3 and 4 with a fixed one-to-one aspect ratio, so that spatial bins render as squares and distances are visually comparable along both axes.

[TODO: Regenerate with a fixed coordinate ratio. Check whether the distortion originated in the plotting call or in the underlying bin geometry; if the bins genuinely are not square, say so explicitly instead of rescaling the display.]

R2.7. Captions for Figures 5 and 6

[TODO]

Figures 5 and 6 require clearer captions, definitions of abbreviations, explanation of rows and columns, and a more explicit connection between the visual evidence and the conclusions drawn.

We have rewritten both captions to be self-contained: each now defines every abbreviation used in the panel, states what the rows and columns represent, and says explicitly what the reader should take from the figure and how it supports the claim made in the text.

[TODO: Rewrite both captions. Figure 5 is the convolution-operator comparison (product, geometric mean, common minimum, with ELSA entrograms) and Figure 6 is the feature-depth sweep producing emergent domains. Each caption must be readable without the surrounding prose.]

R2.8. Controls and baselines

[DONE]

The manuscript would also benefit from additional controls or comparative analyses. At present, there is no clear baseline against which the proposed framework can be evaluated. Possible additions include comparison with existing spatial transcriptomics or ligand-receptor methods, simulated datasets with known ground truth, spatial shuffling controls, ligand-receptor pair shuffling controls, random ligand-receptor combinations, or comparison with previously annotated tissue regions. These analyses would help determine whether the method identifies biologically meaningful spatial structure beyond smoothing, increased feature dimensionality, or clustering.

This was the most important comment in the review, and we have addressed it with two new controls, chosen from the reviewer’s menu: a spatial shuffling control and a comparison with previously annotated tissue regions. Both are reported in a new subsection of the Results and a new supplementary figure, and the code and summary outputs are in the public repository. We also attempted the ligand-receptor pair shuffling control the reviewer suggests, and we report below why we do not present a result from it.

Throughout, spatial coherence of the resulting domain map is quantified with the entropy-based local indicator of spatial association (ELSA) computed on the categorical domain labels, at neighbourhood radii of 50 and 100 units. ELSA is negatively oriented: lower values mean more spatially coherent domains. The domain-generating pipeline is held fixed across observed and control runs, so any difference is attributable to the manipulation alone.

Control A: coordinate shuffle with full model refitting

This is the direct test of whether the spatial smoother manufactures coherent structure from data with no spatial signal. We permuted the assignment of observed counts to spatial locations, destroying spatial structure while preserving the count distribution exactly, and then refit every feature’s spatial model from scratch on the permuted data before running the identical convolution, clustering, and scoring pipeline. This required refitting all 912 features under each of 49 permutations, which we carried out on a high-throughput computing cluster.

The result is unambiguous. The real data yielded 20 domains with ELSA at radius 50 of 0.075. The 49 coordinate-shuffled replicates yielded 4 to 12 domains with ELSA at radius 50 averaging 0.470, ranging from 0.412 to 0.583. Not one of the 49 shuffles came within a factor of five of the observed coherence; the null mean is 6.3 times the observed value. The permutation p -value is 0.020, which is the smallest value attainable with 49 permutations, so the test is limited by the number of replicates rather than by the separation between the observed value and the null.

We regard this as the substantive answer to the reviewer’s concern. Fitting a spatial Gaussian process to spatially meaningless data does not produce coherent tissue domains; it produces a small number of incoherent fragments. The domains we report therefore reflect spatial structure present in the data, not structure imposed by the smoother.

Control B: concordance with independent anatomical annotation

As a positive control we compared the emergent domains to the anatomical annotations published with the source dataset by Chen et al. (2022) for the same tissue section. These annotations were produced independently of our work, by a different method, so they constitute external evidence rather than a circular check.

All 26,000 modeled spatial bins matched an annotated bin. Agreement was substantially above chance: adjusted Rand index 0.401 and normalized mutual information 0.604, against a permutation null whose mean adjusted Rand index was 0.00003 and whose maximum over 499 permutations was 0.001, giving $p = 0.002$. The mean size-weighted purity of the emergent domains with respect to anatomical labels was 77.6%. Individual domains recover recognized structures cleanly: spinal cord at 97% purity, liver at 96%, one heart domain at 99%, meninges at 92%, with lung, muscle, thymus, and cartilage primordium also recovered.

Again we state the result carefully. An adjusted Rand index of 0.401 is substantial but far from perfect agreement, and the disagreement is structured rather than random: our signaling domains subdivide several organs, most visibly the heart, into multiple territories. We interpret this as the expected consequence of measuring a different modality. Anatomical annotation partitions tissue by cell-type composition, whereas our domains partition it by ligand-receptor signaling context, and the two need not coincide. The appropriate conclusion is that the emergent domains recover major anatomical structures while resolving additional substructure, not that they reproduce the anatomical segmentation.

A pair-shuffling control we ran but do not report

We also implemented the ligand-receptor pair shuffling control the reviewer suggests, scrambling the assignment of receptors to ligands while holding the fitted fields, the feature dimensionality, and the clustering procedure fixed. We are not reporting a result from it, and we think the reason is worth stating rather than passing over in silence.

While preparing this revision we checked whether the domain-generating pipeline is deterministic, and found that it is not. The nearest-neighbour graph construction is approximate and multi-threaded, so repeated runs on identical input return somewhat different clusterings. Across eight replicate runs of the identical unshuffled input, ELSA at radius 50 ranged from 0.053 to 0.067, with a standard deviation of 0.0045 and between 19 and 22 domains. The scrambled-pairing null across 99 shuffles had a standard deviation of 0.0053, barely larger. The difference between the true pairing and the scrambled null was about 0.004 in the same units, which is smaller than the run-to-run variability of the pipeline itself.

The pair-shuffling comparison as we ran it therefore cannot separate an effect of the biological pairing from the variability of the clustering algorithm, and any p -value we quoted from it would be misleading. Distinguishing the two would require replicating every condition and comparing distributions rather than single runs. We judged that this particular control is not worth that additional computation, because it is the weakest of the three tests by construction: even a scrambled pairing recombines per-feature fields that are themselves genuinely spatially structured, so it probes only an incremental contribution. The two controls we do report address the reviewer’s concern directly and are not sensitive to this issue, since their effects are far larger than the variability we measured. We note that the underlying variability applies to any single clustering run, including the one shown in Figure 6, and we now say so in the manuscript.

What the controls establish together

Control A shows that the spatial structure is real rather than an artifact of smoothing. Control B shows that the resulting structure corresponds to recognized anatomy, while also resolving substructure within it. Together we believe these address the reviewer’s question of whether the method finds meaningful structure beyond smoothing, dimensionality, and clustering: the coordinate shuffle rules out the smoothing and dimensionality explanations, since the shuffled data pass through an identical pipeline of identical dimensionality, and the anatomical concordance shows the surviving structure is biologically interpretable.

We have not added simulated datasets with known ground truth, the remaining item on the reviewer’s menu. Our reasoning is that a simulation would have to be generated under a spatial model, and any simulation we could write would be close enough to our own fitted model that recovering the truth would demonstrate self-consistency rather than validity. The coordinate-shuffle control tests the same concern using the real data and without that circularity.

R2.9. Reproducibility documentation

[PARTLY DONE]

Reproducibility also needs improvement. Although the repository contains a substantial amount of code, the execution order of scripts, required packages and versions, required input files, intermediate objects, figure-generating scripts, and complete workflow are not sufficiently documented. An external researcher would likely need to reverse engineer the analysis to reproduce the main results.

This criticism was fair and we have acted on it. The repository now documents, for each analysis, the execution order of the scripts, the required input files and where to obtain the public ones, the intermediate objects each stage produces and consumes, and the package versions used.

We have also drawn an explicit line between artifacts that are tracked in the repository and large derived objects that are not. Every untracked derived object is now regenerated by a tracked script,

and the documentation names that script. Where a generating script did not survive, we say so explicitly rather than implying the artifact is reproducible.

[TODO: Only the new controls directory is documented to this standard so far. The main paper pipeline under Supplements for Publication/code/ still needs the same treatment: run order, inputs, intermediates, figure scripts, and a session-info or lockfile capture. This is the largest remaining task after the figures, and the reviewer will check it.]

R2.10. Scientific positioning relative to existing work

[DRAFTED]

The scientific positioning of the manuscript should also be strengthened. The authors should more clearly discuss related work in Gaussian Process-based spatial transcriptomics, Bayesian spatial count modeling, spatial variance component analysis, and spatial ligand-receptor inference. Relevant methods such as SpatialDE, BOOST-GP, Spatial Variance Component Analysis, and SpatialDM should be discussed to clarify which aspects of the manuscript are novel and which represent combinations, extensions, or reinterpretations of existing approaches.

We have added a positioning subsection that discusses each of these method families and states plainly which parts of our contribution are new and which are combinations or reinterpretations of established ideas.

In outline: SpatialDE introduced Gaussian-process testing for spatially variable genes, and we share its use of a GP prior over space but not its goal, since we estimate and use the field itself rather than testing features for spatial variability. BOOST-GP is the closest methodological relative, in that it also places a Gaussian process within a Bayesian count model for spatial transcriptomics; our contribution relative to it is not the count-GP combination, which we do not claim as novel, but the downstream use of the posterior fields as continuous objects that are convolved and then analyzed jointly. Spatial Variance Component Analysis decomposes expression variance into spatial and non-spatial components, which is a variance-partitioning aim rather than a field-estimation one. SpatialDM tests ligand-receptor pairs for spatial association using a bivariate Moran’s statistic, which targets significance of individual interactions where we target a continuous inferred field per interaction with quantified uncertainty.

We have also added a discussion of CytoSignal (Liu et al., *Nature Genetics*, 2026), concurrent work that appeared after our submission and that analyzes the same Stereo-seq mouse embryo dataset. It infers spatially resolved ligand-receptor signaling at cellular resolution using a score with a spatial-permutation null, and validates against experimental proximity-ligation ground truth. We do not claim novelty for spatially resolved ligand-receptor mapping in light of that work, and we have removed language that implied it.

Stating our contribution precisely, and more narrowly than the original submission did: we contribute a generative Bayesian geostatistical treatment of spatialomics data with explicit uncertainty quantification propagated through to the interaction fields, an argument for modeling raw counts as absolute binding density rather than normalized expression, a comparison of convolution operators showing that the common minimum outperforms the product operator conventionally used, and the thesis that clustering the full high-dimensional signaling field without dimensionality reduction produces tissue domains that correspond to anatomy. The combination of a Gaussian process with a Bayesian count likelihood is not itself new, and we now say so.

[TODO: Write this into a new Discussion subsection and add the five citations. Verify the characterization of each method against its paper rather than from summary knowledge, particularly BOOST-GP and SVCA. Confirm the CytoSignal publication details and that the concurrency framing is accurate with respect to both submission timelines.]

R2.11. Generalizability claims

[DRAFTED]

Some claims of generalizability appear broader than the evidence provided. The study is demonstrated using a selected region from a public Stereo-seq mouse embryogenesis dataset. This is useful as a proof of concept, but spatial omics platforms differ in resolution, capture efficiency, sparsity, noise structure, segmentation assumptions, field of view, and biological scale. The authors should moderate broad claims about applicability to most spatial omics datasets or provide additional analyses and practical guidance for different platforms.

We accept this and have moderated the claims rather than attempting to justify them with additional platforms.

The revised manuscript states the scope of the demonstration explicitly wherever a general claim previously appeared: one field of view, from one section, of one embryo, on one platform, at one binning resolution. Claims of the form “applicable to spatial omics data” have been narrowed to describe what we demonstrate, with the extension to other platforms stated as an expectation and its conditions given rather than asserted.

We have added a short paragraph on what would need to hold for the approach to transfer, which we think is more useful to a reader than a broad claim: the observations must be counts with an interpretable exposure or offset, the spatial support must be dense enough that a Gaussian process is identifiable at the length scales of interest, and at single-cell-resolution platforms the binning step we rely on would need to be replaced by an explicit treatment of cell geometry. We also note that platforms with substantially sparser capture may favour a zero-inflated likelihood, which our Table 1 comparison found unnecessary at this sparsity but which the framework accommodates.

[TODO: Apply the softening pass across the manuscript. The Abstract, the end of the Introduction, and the Discussion are where the broad claims live. Check that the moderated scope is consistent everywhere, including the Abstract.]

R2.12. Caution in biological interpretation

[DRAFTED]

Finally, the biological interpretation should remain cautious. I appreciate that the authors acknowledge that transcript abundance is only used as a proxy for protein abundance. However, because downstream analyses involve inferred ligand-receptor interactions and morphogenic signaling fields, the manuscript should avoid language that implies direct evidence of protein-level interactions, ligand-receptor binding, signaling activity, or information transmission. Terms such as “proxy,” “inferred field,” “model-derived estimate,” or “potential interaction” would be more appropriate.

We agree, and we have revised the language throughout.

We have replaced formulations that implied observed protein-level events with formulations describing model-derived estimates from transcript measurements. Descriptions of ligand-receptor quantities now consistently use “inferred interaction field” or “potential interaction” rather than

wording that implies binding was observed; references to signaling activity are now phrased as transcriptional evidence consistent with potential signaling; and language implying information transmission between regions has been removed or explicitly marked as interpretation.

We have also added an explicit statement of the inferential chain and its assumptions in the Discussion, so a reader can see exactly how far the claims are from the measurement: transcript counts are a proxy for transcript abundance, transcript abundance is a proxy for protein abundance, co-localization of ligand and receptor proxies is a proxy for the opportunity for interaction, and none of these steps is directly observed here.

[TODO: Perform the terminology pass across the full manuscript. Read the Results and Discussion specifically for verbs implying observation of binding or signaling. Keep terminology consistent: pick one term per concept and use it throughout.]

6. Additional corrections

Internal note: the item below was found by us, not by a reviewer. It should still be disclosed in the letter, since the reviewer will notice the numbers changed.

In preparing this revision we audited the feature and interaction counts reported in the original manuscript against the analysis outputs, and found two errors that we have corrected throughout. We report them here because they change numbers that appear in the text, in Table 1, and in Figure 6.

First, the attrition from the curated feature list was not described. The list contains 918 ligand or receptor genes. Three (*a*, *Calcb*, *Cckar*) were excluded before fitting, and three more (*Fgf1*, *Fzd7*, *Trhr*) failed to converge, leaving 912 features with fitted spatial fields. The original text stated that the spatial fields of all 918 features were modeled. The revised text gives the correct figure and explains the attrition.

Second, the number of interaction fields was overstated by four. The manuscript reports 1481 unique ligand-receptor interaction fields, and describes them as following from the 918-gene list. On inspection, the 1481-column object used for the high-dimensional clustering contains four pairs of exactly duplicated columns, which we traced to files that had been silently duplicated by a cloud file-synchronization conflict and then read in alongside their originals. The four affected interactions are PDGFC-PDGFR α , PDGFC-PDGFR β , PDGFD-PDGFR α , and PDGFD-PDGFR β ; in each case the duplicate column is numerically identical to its counterpart. The correct number of distinct interaction fields computable from the 912 fitted features is therefore 1477. We have corrected the count wherever it appears, including the upper limit of the feature-depth sweep in Figure 6.

We do not think this materially affects the results. The duplication gives four of 1477 interactions twice their intended weight in a scaled clustering over more than a thousand dimensions, and it applies equally to every point in the analysis, so it cannot create spatial structure. We have not attempted to quantify the difference by re-running the clustering without the duplicates, because the clustering algorithm is not deterministic (see our response to R2.8), so a single re-run would not isolate an effect of this size from run-to-run variation.

[TODO: Apply these corrections to every occurrence in the manuscript and captions: 918 to 912 at line 217; 1481 to 1477 at lines 213 and 251 and in the Fig 6 caption; and the Table 1 caption, which averages over the fitted features that pass the score filter, not over “all 918 modeled ligand/receptor features.” Decide with Raredon whether

to name the synchronization conflict as the cause in the manuscript or to describe it more neutrally as duplicated input files.]